AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 14 · STUDENT LESSON

Learning rate experiments

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Restart the same state. Change one rate. Keep every curve.

Explain rate-dependent paths with complete controlled parameter and loss evidence.

Keep one habit: Full protocol → five complete trajectories and graphs → pictured matching endpoints → bound/scaling/control probes → independent new objective.

MAIN LESSON · SECTION 1

Did only the learning rate change?

> Restart the same state and retain every parameter and loss value.

Learning goals

I can run a **controlled comparison** of **learning rate**s.

I can explain slow progress **oscillation** and increasing loss in a toy model.

I can report the limits of a **training** experiment.

Retrieve update timing and budgets

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Batching and epochs. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Updating parameters. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Loss functions. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Why can equal epochs give different update counts when batch size changes?

> **My starting explanation**
>
>
>
>
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End goal

By the end, I can explain **learning rate** experiments using a traced example and a stated limitation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Run a **controlled comparison** of **learning rate**s

Explain slow progress **oscillation** and increasing loss in a toy model

Report the limits of a **training** experiment

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Run five rates under one fully specified protocol

> Same two training pairs,same mean objective,same start and four full-data updates.

Complete restricted-network experiment

Use the same restricted network h = ReLU(x), y = vh, with only v trained. Keep the fictional **training** pairs (1, 2) and (2, 4), start v = 0, and use one full-data batch per update.

Compare rates 0.05, 0.2, 0.3, 0.4 and 0.5 for exactly four updates each. Restart at zero for every run.

The **training** objective is ordinary MSE: L(v) = 2.5 times (v - 2) squared. Its gradient is 5 times (v - 2). A step is new v = v - rate times 5 times (v - 2).

This small quadratic lets us explain each result rather than treating the rate as a magic setting.

Complete first/fourth-state comparison

Rate	v after first update	First training MSE	v after four updates
0.05	0.5	5.625	1.3671875
0.2	2	0	2

Rate	v after first update	First training MSE	v after four updates
0.3	3	2.5	1.875
0.4	4	10	0
0.5	5	22.5	-8.125

Keep the one-step optimum scoped to this quadratic

The same rate need not behave this way for another model, **input** scale, loss convention or batch rule. Rate 0.2 is an exact one-step solution here because of this particular gradient formula.

Name supplied and missing facts

> **List the given facts and one fact that is not supplied.**

MAIN LESSON · SECTION 3

Read complete paths rather than only the final score

> A sign-changing distance can shrink,stay the same or grow.

Derive the signed-distance recurrence

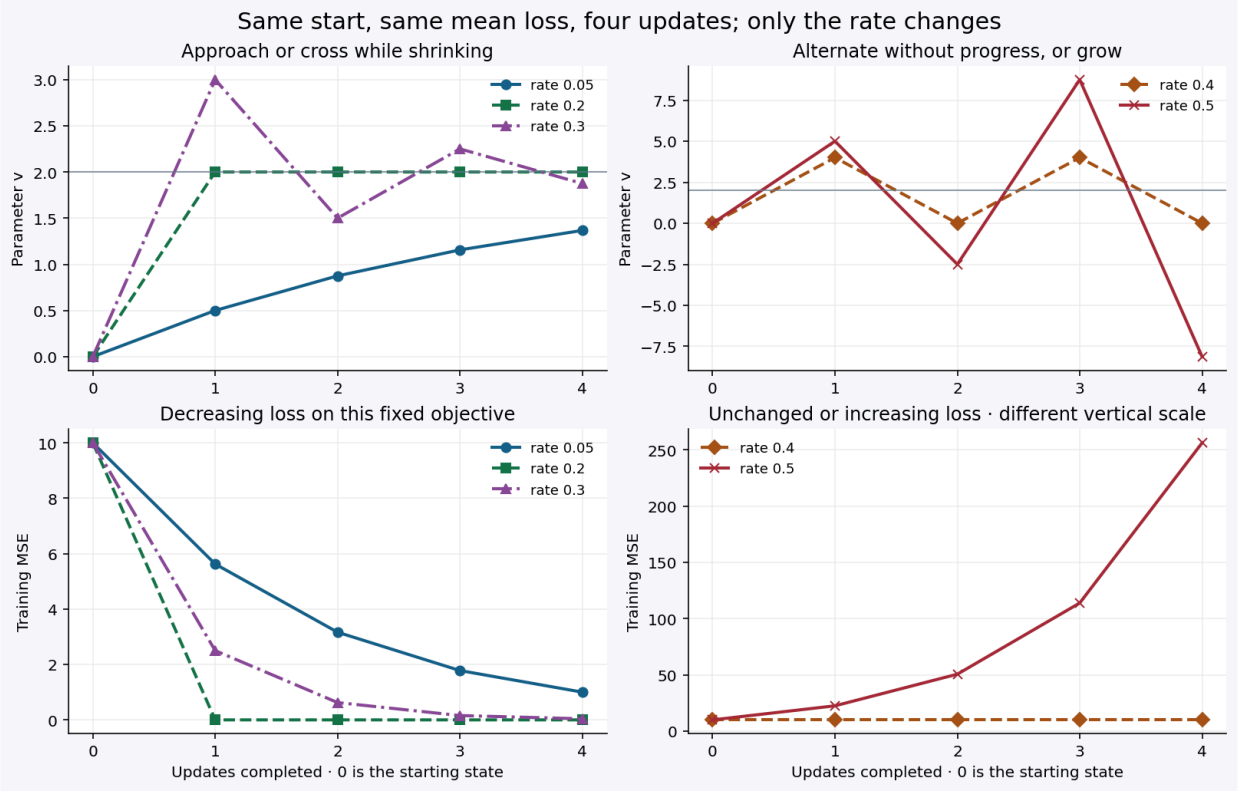
Write d = v - 2, the signed distance from this objective minimum. Substituting the update gives new d = (1 - 5 times rate) times old d. At rate 0.05 the factor is 0.75: distance shrinks without changing sign.

At rate 0.3 it is -0.5: distance changes sign and halves. Both can reduce loss.

Every logged parameter and loss state

Rate	All five parameter states(0–4 updates)	All five training MSEs	Distance multiplier 1–5 rate
0.05	[0.0, 0.5, 0.875, 1.15625, 1.3671875]	[10.0, 5.625, 3.1640625, 1.77978515625, 1.001129150390625]	0.75
0.2	[0.0, 2.0, 2.0, 2.0, 2.0]	[10.0, 0.0, 0.0, 0.0, 0.0]	0.0
0.3	[0.0, 3.0, 1.5, 2.25, 1.875]	[10.0, 2.5, 0.625, 0.15625, 0.0390625]	-0.5
0.4	[0.0, 4.0, 0.0, 4.0, 0.0]	[10.0, 10.0, 10.0, 10.0, 10.0]	-1.0
0.5	[0.0, 5.0, -2.5, 8.75, -8.125]	[10.0, 22.5, 50.625, 113.90625, 256.2890625]	-1.5

Labelled controlled parameter/loss curves



Curve provenance and helper contract

The saved curves are computed from the exact declared update v←v−rate×5(v−2), starting 0 and logging states 0 through4. They are reproducible toy calculations, not measurements from a real neural training service. Write d=v−2:each step multiplies d by1−5 rate and multiplies loss by its square. Crossing sides can still shrink the distance.

The numerical helper assumes finite rate and a nonnegative integer update count; it does not validate them or choose a rate for an unseen task.

The helper assumes a finite rate and nonnegative integer update count. Its five logged states include the initial value.

At 0.3 the parameter sequence is [0, 3, 1.5, 2.25, 1.875] and loss is [10, 2.5, 0.625, 0.15625, 0.0390625]. Crossing the minimum does not itself mean **training** failed.

Complete runnable rate experiment

```
def run(rate, updates=4):
    v = 0.0
    history = [v]
    for _ in range(updates):
        gradient = 5 * (v - 2)
        v = v - rate * gradient
        history.append(v)
    losses = [2.5 * (value - 2) ** 2 for value in history]
    return history, losses

for rate in [0.05, 0.2, 0.3, 0.4, 0.5]:
    values, losses = run(rate)
    print(rate, values, losses)
```

Connect each curve to the update rule

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Compare rates with the same start and budget

> Distinguish a first crossing from repeated divergence.

Current crossing task

- At rate 0.3 the first update gives v = 3, past the minimum at 2. Calculate the next v and loss. Has the first crossing proved **divergence**?

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the pictured rate1restart

- Change: Add a rate 1 run from a = 0.

▼ **Check after predicting**

Prediction: Its path is 8 then 0, with loss 16 at both steps.

Reason: The update alternates across 4 without reducing the initial loss.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

See different paths with the same final loss

> The comparison needs its objective,start and exact update budget.

Complete pictured objective and separate scope

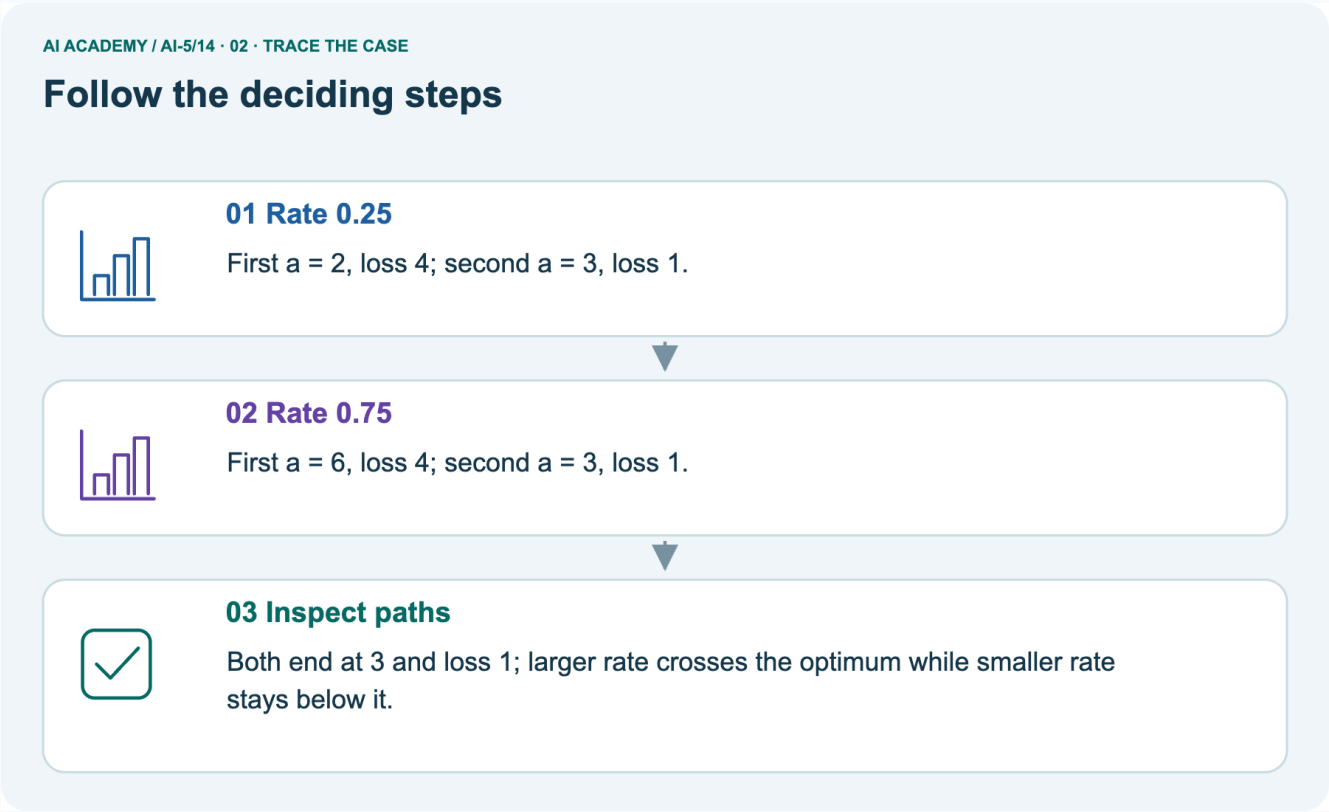
Compare **learning rate**s with identical starting loss and update counts.

Loss $J(a)=(a-4)^2$; gradient $2(a-4)$; start $a = 0$. Compare two updates at rates 0.25 and 0.75.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture uses a different objective $J(a)=(a-4)^2$, gradient $2(a-4)$, start 0 and two updates. Rate.25 gives a2 then3, loss 4 then1; rate.75 gives a6 then3, also loss 4 then1. Different paths happen to share the final result. A separate restarted rate 1 run gives a8 then0, loss 16 both times. These rates belong to that objective, not the main $2.5(v-2)^2$. The independent task supplies a third objective centred at1.

Trace both two-update paths



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Rate 0.25	First a = 2, loss 4; second a = 3, loss 1.
2. Rate 0.75	First a = 6, loss 4; second a = 3, loss 1.
3. Inspect paths	Both end at 3 and loss 1; larger rate crosses the optimum while smaller rate stays below it.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Keep the starting parameter and update budget identical to compare rates fairly. Rate one-quarter approaches the optimum from below; rate three-quarters crosses it and comes back.

Their final losses happen to match after two updates, despite different paths. Rate one repeatedly overshoots by the same distance, so using the correct gradient alone does not ensure improvement.

Predict the added rate1run

- Change one thing: Add a rate 1 run from a = 0. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting

AI ACADEMY / AI-5/14 · 04 · COMPARE THE CASES

Change one thing. Trace what follows.

Original case

Both end at 3 and loss 1; larger rate crosses the optimum while smaller rate stays below it.

Changed condition

Add a rate 1 run from a = 0.

Worked changed-case result

Its path is 8 then 0, with loss 16 at both steps.
Why: The update alternates across 4 without reducing the initial loss.

Follow the labelled picture with the matching explanation. Open the vector diagram

Explain fair controls and model-specific limits

- These quadratic paths explain step size. Real nonconvex networks can have more complicated rate behaviour.

- Read the labelled picture: What should a fair rate comparison hold fixed? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Try the main rate.4oscillation

The update factor explains three behaviours

RATE AND FACTOR	SIGNED DISTANCE BEHAVIOUR	TRAINING LOSS BEHAVIOUR
0.05 gives 0.75	Same side and shrinking	Multiplied by 0.5625
0.3 gives -0.5	Alternating and shrinking	Multiplied by 0.25
0.5 gives -1.5	Alternating and growing	Multiplied by 2.25

For this objective, loss is 2.5 times squared distance. Therefore the loss multiplier is the square of the distance multiplier.

For rate 0.4, calculate its distance multiplier, the first two new v values and both losses. Does a changing parameter guarantee progress?

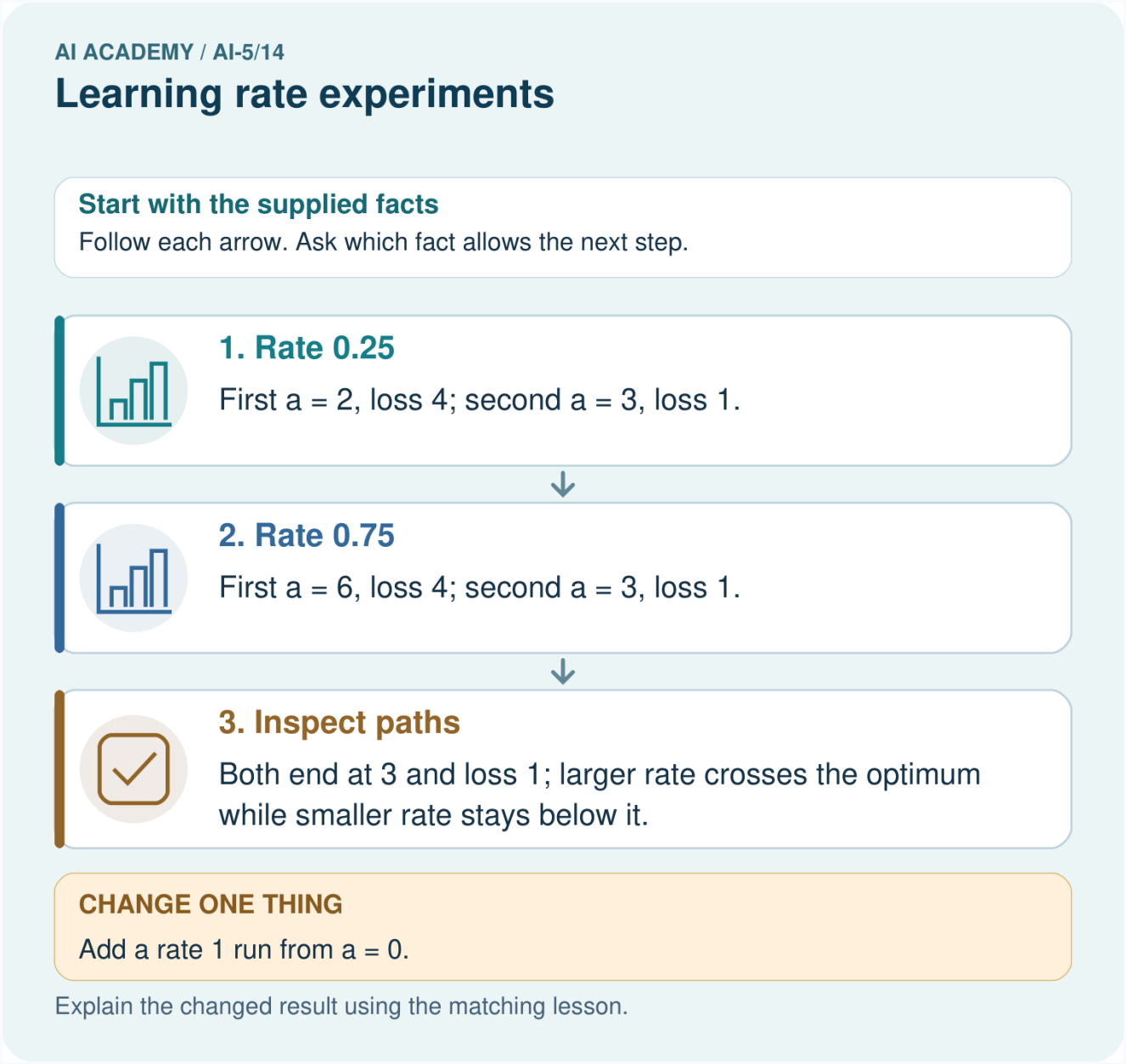
▼ **Earlier wording, alternatives and picture transcripts**

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Learning rate experiments

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Repair universal rate rules and changed-objective comparisons

> A rate is meaningful together with the gradient scale and update definition.

Exact quadratic bound and mean-versus-sum scaling

For a nonzero starting distance, loss strictly decreases on each step of this quadratic when the absolute value of 1 - 5 times rate is below 1. For positive rates this means 0 < rate < 0.4. At 0 the model does not move.

At 0.4 it oscillates with unchanged loss. Above 0.4 its distance grows. These bounds belong to this model and objective, not to neural networks generally.

If we switch from mean squared error to the sum over the same two examples, the gradient doubles from 5(v - 2) to 10(v - 2). Keeping the old rate changes the update. To reproduce the original steps, halve the rate.

Likewise, rescaling features can change gradient magnitudes. Record the loss and units alongside the rate.

Find the sum-objective equivalent rate

- Replace MSE by the sum of squared errors on the two **training** pairs. Which rate reproduces the MSE run at rate 0.3? Explain using gradients.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise trajectory reasoning with less help

> State the factor,its sign and whether its magnitude shrinks.

Current task and response

- For rate 0.05, start v = 0 and compute two updates and their **training** MSE values using the lesson objective.

> **For rate 0.4, calculate its distance multiplier, the first two new v values and both losses. Does a changing parameter guarantee progress?**

> **Explain your prediction to a partner before comparing results.**

Supported W2record

Use workbook W2 for the connected example “**Learning rate** experiments”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual misconception check

- Use the worked case for **Learning rate** experiments. Proposed explanation: “Crossing the minimum means **divergence**”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Pictured fair-comparison reasoning

- What should a fair rate comparison hold fixed? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**
>
> _____
>
> _____
>
> _____
>
> _____

Reserve the independent new objective

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

Practise with less help: Explain a crossing

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run the complete five-rate comparison and control audit

> All settings,paths and expected boundary behavior are supplied.

Current crossing task and complete runnable experiment

At rate 0.3 the first update gives v = 3, past the minimum at 2. Calculate the next v and loss.
Has the first crossing proved **divergence**?

```
def run(rate, updates=4):
    v = 0.0
    history = [v]
    for _ in range(updates):
        gradient = 5 * (v - 2)
        v = v - rate * gradient
        history.append(v)
    losses = [2.5 * (value - 2) ** 2 for value in history]
    return history, losses

for rate in [0.05, 0.2, 0.3, 0.4, 0.5]:
    values, losses = run(rate)
    print(rate, values, losses)
```

Complete normal,boundary,missing,scaling and confounding probes

Complete controlled-rate probe	Expected result / action
Normal rate.3 after first crossing v3	Next gradient 5, next v1.5, loss.625; crossing did not prove divergence
Boundary rate.4	Multiplier−1; v4 then0, loss 10 then10
No-change rate 0	All five states v0, loss 10
Fast exact toy rate.2	First v2, then stays; loss 0 from first update
Missing rateNoneorinvalid update count	Outside finite-rate/nonnegative-integer contract; validate before calling
Change MSE to two-row SSE	Gradient doubles; halve rate.3to.15 for identical steps
Change rate and batch size/start together	Cannot isolate rate; restart same original state and hold all other settings fixed
Choose from final-test outcomes	Those cases become selection/development evidence; retain a fresh frozen final gate

Predict,restart,run or trace,and retain every path

1. Predict all five recorded paths before running or tracing the supplied code. Restart v0 for each rate; retain every loss, not only a preferred endpoint. Compare the returned values with the complete table and graph, including their different vertical scales. Trace the zero/boundary and mean-to-sum cases separately. The helper assumes valid configuration and has no missing-input parser.

   Report these two-row training results as a controlled classroom experiment; use separate validation for development choices and keep final-test references out of rate selection.

2. On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Keep the evidence ledger

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: compare rates on a different quadratic

> Use its own gradient scale,start and two-update budget.

Complete new independent objective and rates

Use J(a) = (a - 1) squared with gradient 2(a - 1), start a = 0 and two updates. Restart for each rate 0.25, 0.5 and 1. Compute the two new a values and losses for each run. Identify the result

supported after two updates.

First attempt and support record

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Compare a new objective independently

Attempt this without hints

Use J(a) = (a - 1) squared with gradient 2(a - 1), start a = 0 and two updates. Restart for each rate 0.25, 0.5 and 1. Compute the two new a values and losses for each run. Identify the result supported after two updates.

> **My independent procedure, result and limitation**

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the rate with its full experimental protocol

> A reader should know what stayed fixed and what the curves measure.

Project artifact and evidence table

Add a labelled **learning rate** experiments **evidence** sheet to Image-pattern Network Investigation. Show how you can run a **controlled comparison** of **learning rate**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run/trace and limits

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ Earlier wording, alternatives and picture transcripts

Evidence for your project

MAIN LESSON · SECTION 16

Finish with one controlled comparison and one bounded conclusion

> Distinguish the best observed training result from a universal setting.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Run a **controlled comparison** of **learning rate**s

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use the mathematical words precisely**

REFERENCE · SECTION 5

Reference · Use the mathematical words precisely

> Connect each term to a value or step in the current example.

Vocabulary and your explanation

Word	Meaning
Learning rate	The multiplier controlling the size of a gradient update
Controlled comparison	An experiment changing a named factor while holding other stated conditions fixed
Oscillation	Repeated movement from one side of a value to the other
Divergence	Movement away from a target solution with growing error in this experiment

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- For rate 0.05, start v = 0 and compute two updates and their **training** MSE values using the lesson objective.
- Replace MSE by the sum of squared errors on the two **training** pairs. Which rate reproduces the MSE run at rate 0.3? Explain using gradients.
- Run A starts v = 0 with full-data batches for four updates. Run B starts at the final A value, changes both rate and batch size, then compares final **training** losses. Can the difference be attributed only to rate?

  Propose a repair.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the representation and its limits

REFERENCE · SECTION 12

Reference · Check the representation and its limits

> Use after practice;reserve the new independent $J(a)=(a-1)^2$two-update rate comparison for the first individual attempt.

Current explanation and response

- Record the initial state, data, frozen values, objective, batch rule, update budget, candidate rates, parameter traces and loss traces. These runs compare **training** behaviour on two examples.

  They do not establish which setting performs best on independent data. Keep the final test set out of rate selection; later lessons use validation curves to make development choices.

- Next week compares **training** and validation curves, including when their trends disagree.

- Run A starts v = 0 with full-data batches for four updates. Run B starts at the final A value, changes both rate and batch size, then compares final **training** losses. Can the difference be attributed only to rate?

  Propose a repair.

> **A team says rate 0.2 is the best rate for every neural network because it reached zero training loss here. Replace this claim with a precise experimental conclusion and one limitation.**

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

The update factor explains three behaviours

RATE AND FACTOR	SIGNED DISTANCE BEHAVIOUR	TRAINING LOSS BEHAVIOUR
0.05 gives 0.75	Same side and shrinking	Multiplied by 0.5625
0.3 gives -0.5	Alternating and shrinking	Multiplied by 0.25
0.5 gives -1.5	Alternating and growing	Multiplied by 2.25

For this objective, loss is 2.5 times squared distance. Therefore the loss multiplier is the square of the distance multiplier.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ **Reference · The earlier route checklist**

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Compare **learning rate**s with identical starting loss and update counts.

By the end, I can	How I will show it
Run a controlled comparison of learning rates	Explain the deciding step, show a trace or test, and state one limit.
Explain slow progress oscillation and increasing loss in a toy model	Explain the deciding step, show a trace or test, and state one limit.
Report the limits of a training experiment	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Compare **learning rate**s with identical starting loss and update counts.
- Which labelled step matches this detail? Both end at 3 and loss 1; larger rate crosses the optimum while smaller rate stays below it.
- 1. Inspect paths
- 2. Rate 0.25
- 3. Rate 0.75
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Inspect paths Both end at 3 and loss 1; larger rate crosses the optimum while smaller rate stays below it.

Keep the starting parameter and update budget identical to compare rates fairly. Rate one-quarter approaches the optimum from below; rate three-quarters crosses it and comes back.

Their final losses happen to match after two updates, despite different paths. Rate one repeatedly overshoots by the same distance, so using the correct gradient alone does not ensure improvement.

Changed-case question and optional hint

- Changed-case practice: Add a rate 1 run from a = 0. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. Its path is 8 then 0, with loss 16 at both steps.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Its path is 8 then 0, with loss 16 at both steps. The update alternates across 4 without reducing the initial loss.

Repair and retain the record

Review Batching and epochs (ai-5/13). Goal: Distinguish an example a batch an update and an epoch. Short practice: Why do the single-example orders finish at different weights?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Updating parameters (ai-5/11). Goal: Apply a **learning rate** to an ordered gradient. Short practice: Why must new loss be recomputed rather than assumed smaller?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Loss functions (ai-5/8). Goal: Calculate per-example squared error and mean squared error. Short practice: Why must the report name the chosen loss?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

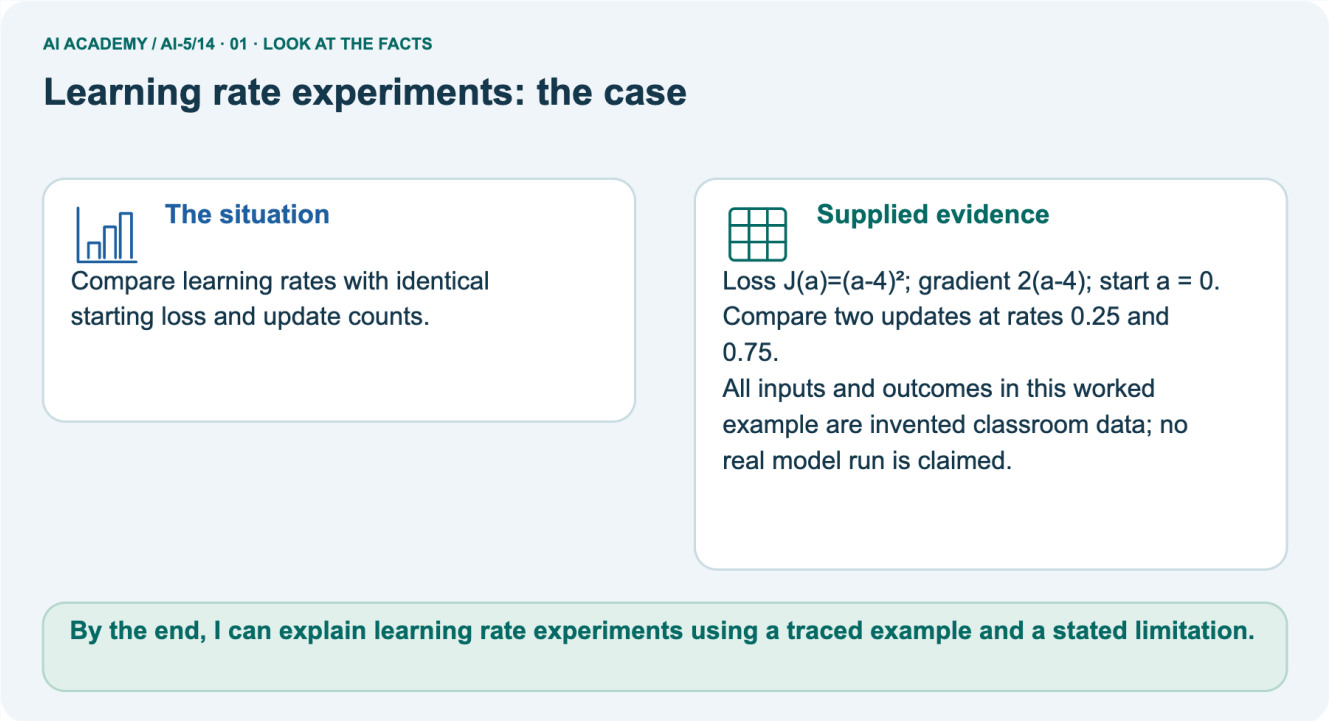
▼ Reference · Six-picture concept journey

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Learning rate experiments: the case



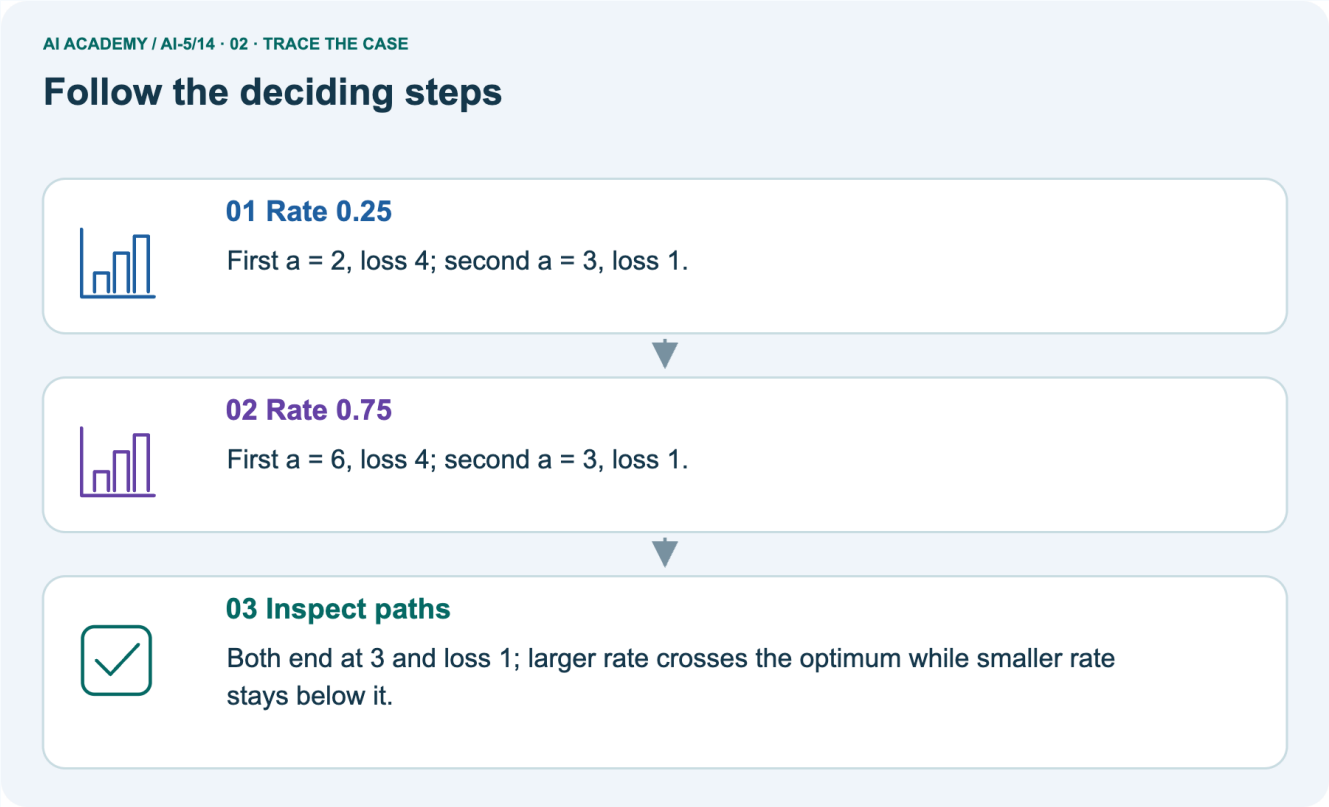
Follow the labelled picture with the matching explanation. Open the vector diagram

Compare **learning rate**s with identical starting loss and update counts. Supplied facts: Loss J(a)=(a-4)²; gradient 2(a-4); start a = 0. Compare two updates at rates 0.25 and 0.75.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

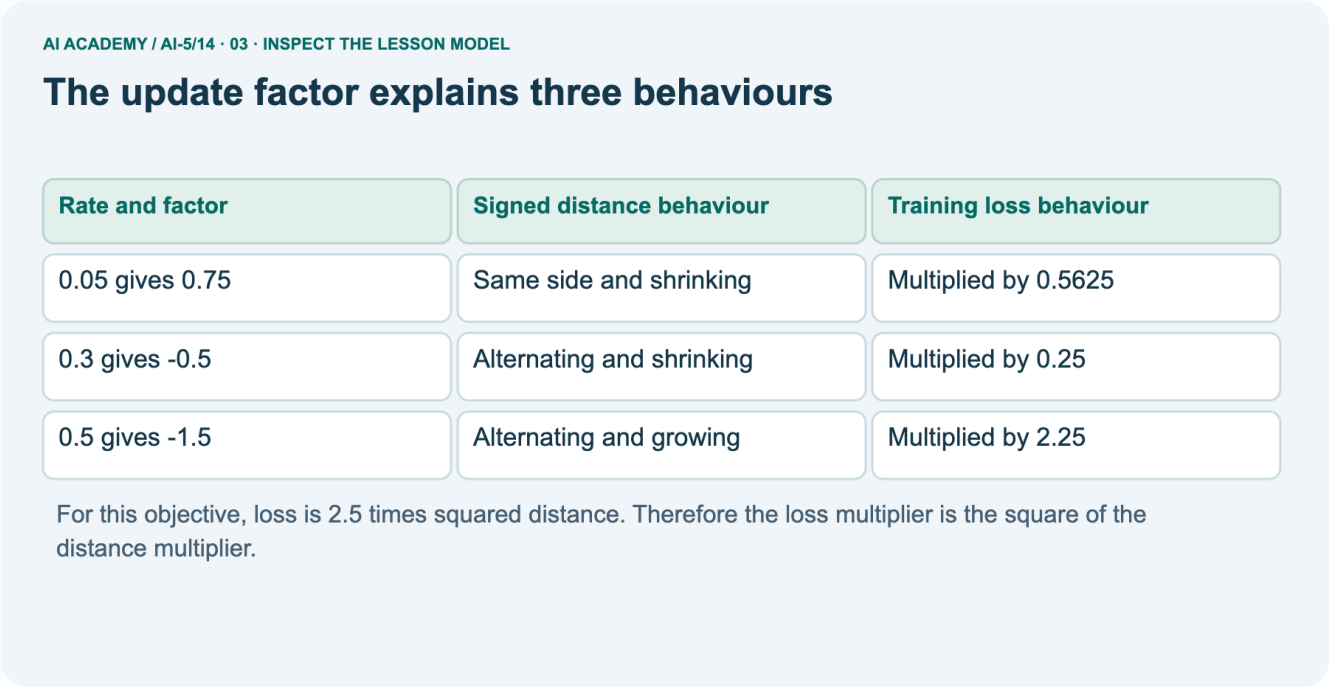


Follow the labelled picture with the matching explanation. Open the vector diagram

Rate 0.25: First a = 2, loss 4; second a = 3, loss 1. → Rate 0.75: First a = 6, loss 4; second a = 3, loss 1. → Inspect paths: Both end at 3 and loss 1; larger rate crosses the optimum while smaller rate stays below it.

Point to each numbered stage and explain which recorded **input** it uses.

03. The update factor explains three behaviours



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The update factor explains three behaviours

Rate and factor	Signed distance behaviour	Training loss behaviour
0.05 gives 0.75	Same side and shrinking	Multiplied by 0.5625
0.3 gives -0.5	Alternating and shrinking	Multiplied by 0.25
0.5 gives -1.5	Alternating and growing	Multiplied by 2.25

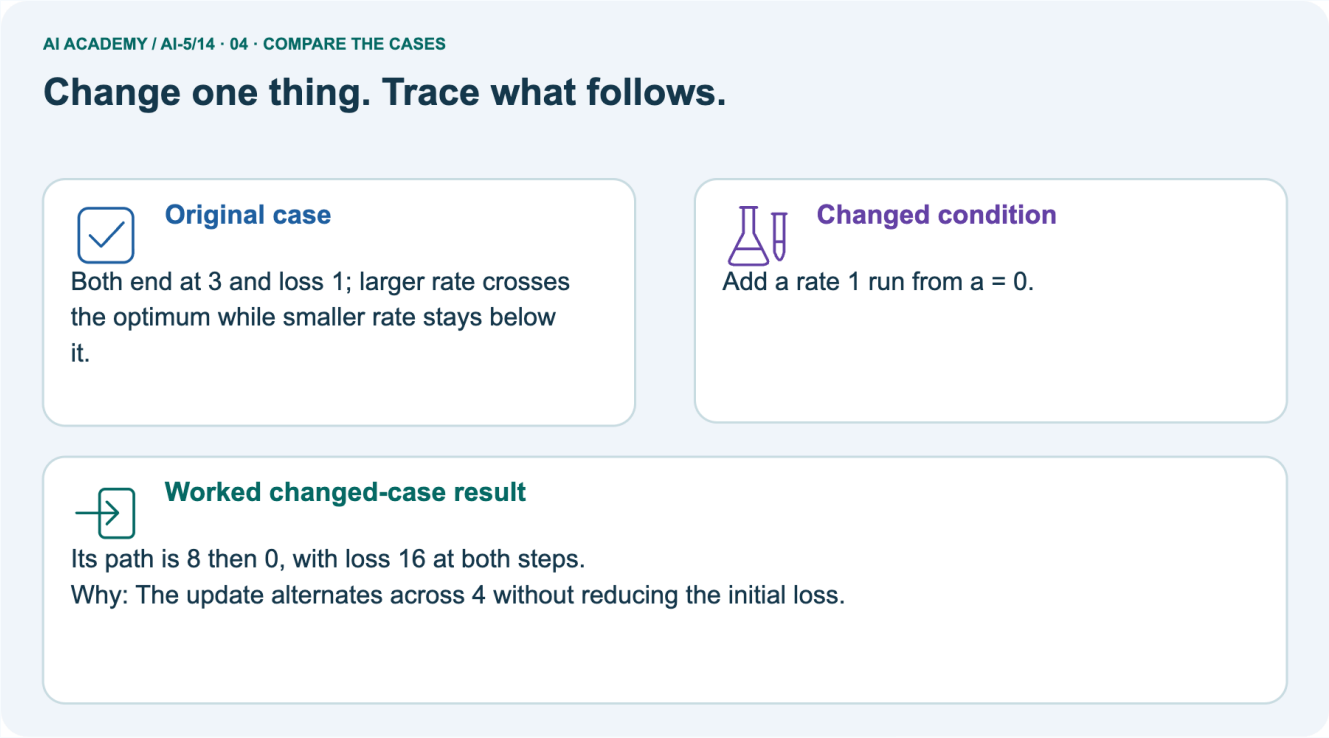
For this objective, loss is 2.5 times squared distance. Therefore the loss multiplier is the square of the distance multiplier.

Follow the labelled picture with the matching explanation. Open the vector diagram

The update factor explains three behaviours. 0.05 gives 0.75 | Same side and shrinking | Multiplied by 0.5625. 0.3 gives -0.5 | Alternating and shrinking | Multiplied by 0.25. 0.5 gives -1.5 | Alternating and growing | Multiplied by 2.25

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



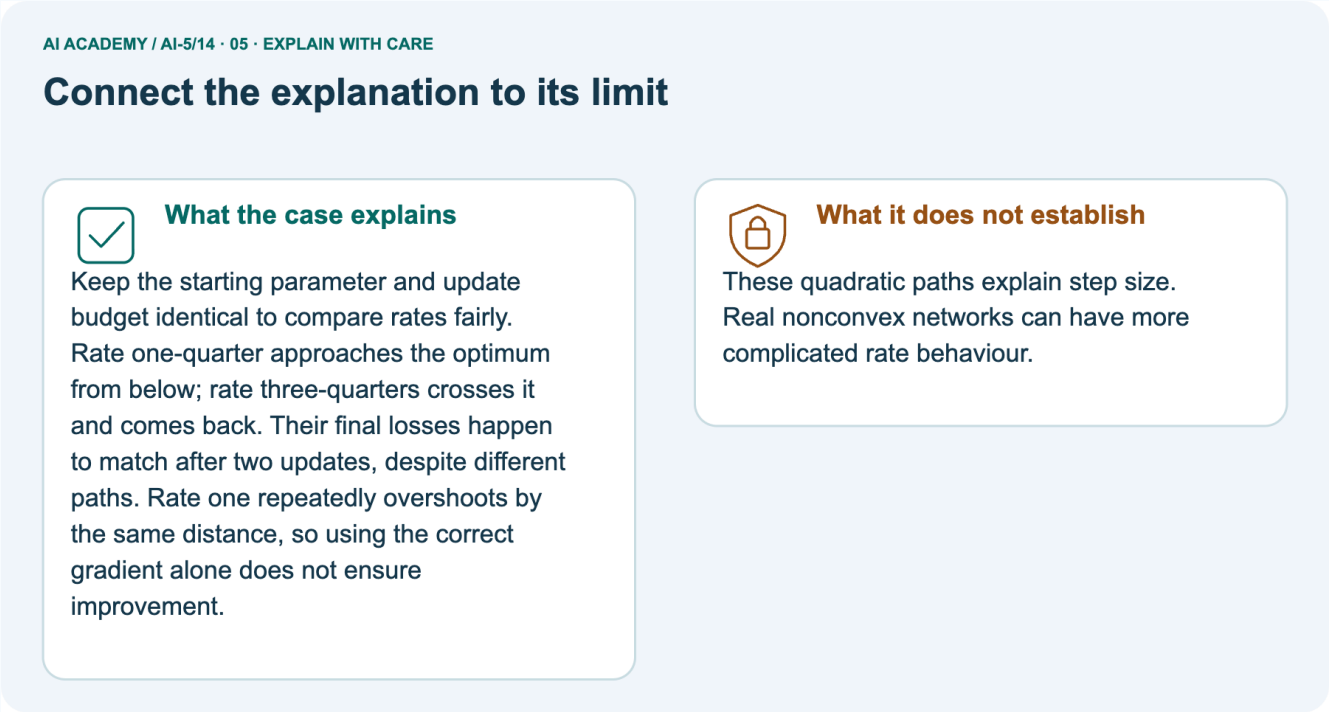
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Both end at 3 and loss 1; larger rate crosses the optimum while smaller rate stays below it. Changed condition: Add a rate 1 run from a = 0. Expected result: Its path is 8 then 0, with loss 16 at both steps.

Reason: The update alternates across 4 without reducing the initial loss.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

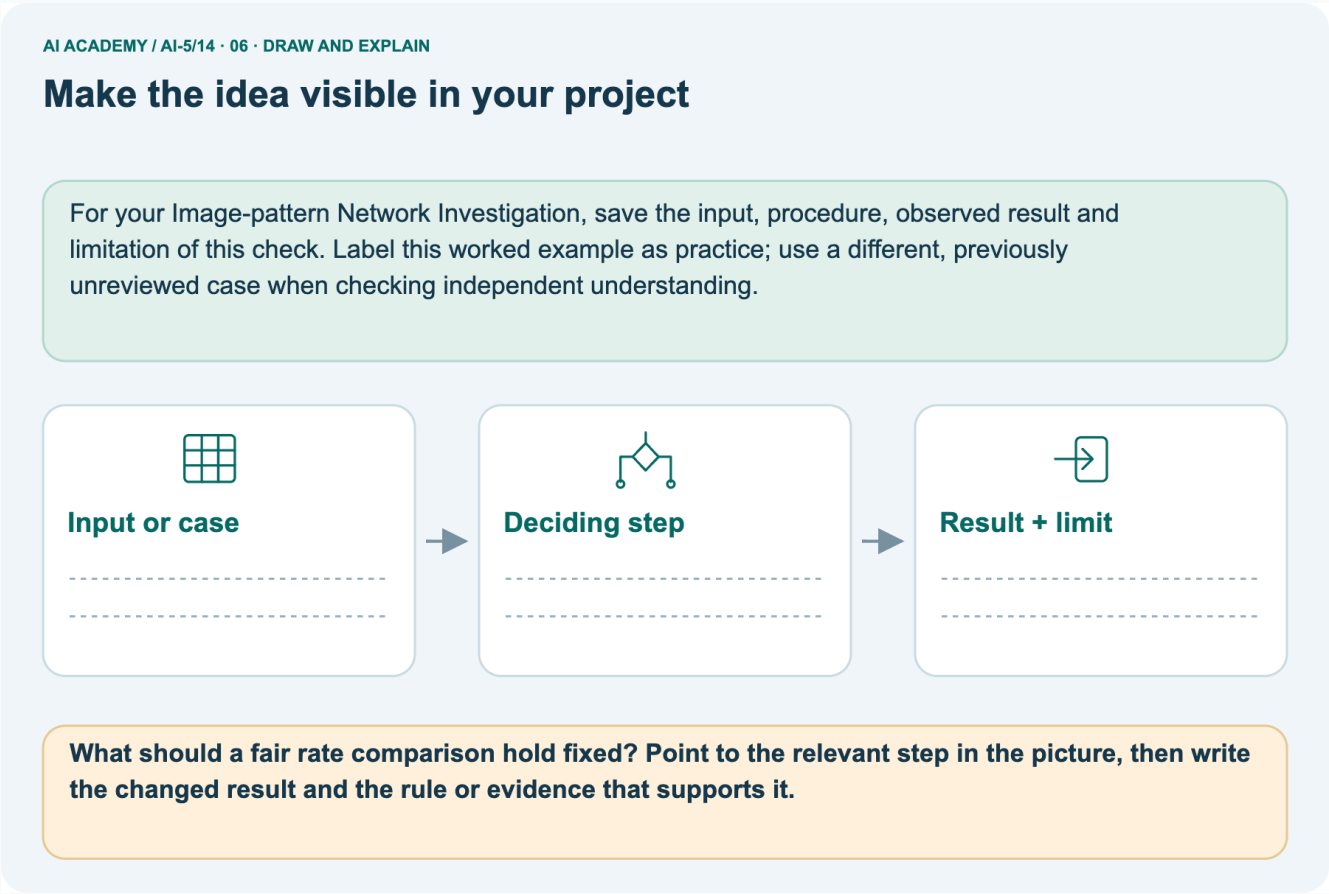
Keep the starting parameter and update budget identical to compare rates fairly. Rate one-quarter approaches the optimum from below; rate three-quarters crosses it and comes back.

Their final losses happen to match after two updates, despite different paths. Rate one repeatedly overshoots by the same distance, so using the correct gradient alone does not ensure improvement.

Limit: These quadratic paths explain step size. Real nonconvex networks can have more complicated rate behaviour.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What should a fair rate comparison hold fixed? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Learning rate experiments: the case

02. Follow the deciding steps

03. The update factor explains three behaviours

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.